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Unit – 5 : Session – 11 : SLO – 1 | Memory-Mapped vs Port-Mapped I/O

■ I. Multiple Choice

1. B) Memory locations — MMIO treats device registers as memory addresses.
2. B) x86 CISC — port-mapped (isolated) I/O is characteristic of x86.
3. C) IN/OUT — the special instructions used for port-mapped I/O.

■ II. Fill in the Blanks

1. memory — MMIO shares its address space with main memory.
2. IN/OUT — PMIO uses special instructions such as IN and OUT.
3. RISC — MMIO is commonly used in RISC processors (ARM, MIPS).

■ III. True or False

1. TRUE — MMIO uses ordinary LOAD/STORE instructions.
2. FALSE — PMIO has a separate port address space.
3. TRUE.

■ IV. Match the Following

- A) MMIO — 2) Uses memory addresses.
B) PMIO — 1) Uses ports.
C) LOAD — 3) Read from memory.
D) IN — 4) Read from port.

■ V. Numerical Answers

1. 0x1004 — base 0x1000 + offset 4.
2. 50 — the polling loop checks the status 50 times.
3. 256 — an 8-bit port address gives $2^8 = 256$ ports.